NAME: _____ Geometry (FL B.E.S.T.)

10.2 Volume

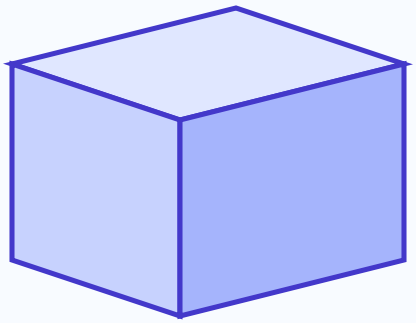
Volume measures the space enclosed by a solid, in cubic units. Prisms and cylinders use $V = Bh$ (base area times height); pyramids and cones use exactly $\frac{1}{3}$ of that, $V = \frac{1}{3}Bh$, because a pyramid/cone fills only a third of the prism/cylinder that encloses it; a sphere uses $V = \frac{4}{3}\pi r^3$. As with surface area, the height needed in the formula is the solid's *true vertical height*, which sometimes must be recovered from a slant height using the Pythagorean Theorem. This worksheet also covers composite solids, cone-vs-cylinder comparisons, reverse problems, and real-world capacity/unit-conversion problems.

Examples

Example 1: Rectangular prism

Find the volume of a rectangular prism with length 8, width 5, height 6.

Solution



Volume of a box = length × width × height.

Identify the dimensions: $l = 8$, $w = 5$, $h = 6$.

Apply the box volume formula:

$V = lwh = 8(5)(6) = 240$.

Answer: 240 cubic units.

Example 2: Cube

Find the volume of a cube with edge length 7.

Solution

Identify the edge length: $s = 7$.

A cube's volume is its edge length cubed:

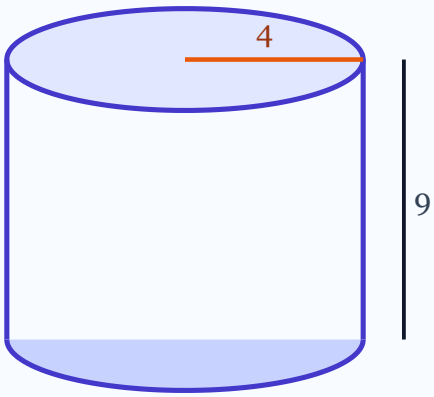
$V = s^3 = 7^3 = 343$.

Answer: 343 cubic units.

Example 3: Cylinder

Find the volume of a cylinder with radius 4, height 9.

Solution



Base area πr^2 times height.

Identify the radius and height: $r = 4$, $h = 9$.

Apply the cylinder volume formula:

$V = \pi r^2 h = \pi(16)(9) = 144\pi$.

Answer: 144π cubic units.

Example 4: Square pyramid — height given

A square pyramid has base edge 6 and height 10. Find its volume.

Solution

Base area: $B = 6^2 = 36$.

$$V = \frac{1}{3}Bh = \frac{1}{3}(36)(10) = \frac{360}{3} = 120.$$

Answer: 120 cubic units.

Example 5: Square pyramid — find the height first

A square pyramid has base edge 16 and slant height 17. Find its volume.

Solution

The formula needs the vertical height h, not the slant height $\ell = 17$. Since ℓ is the hypotenuse of a right triangle with legs h and half the base edge:

$$h = \sqrt{\ell^2 - \left(\frac{s}{2}\right)^2} = \sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15.$$

Now find the base area: $B = 16^2 = 256$.

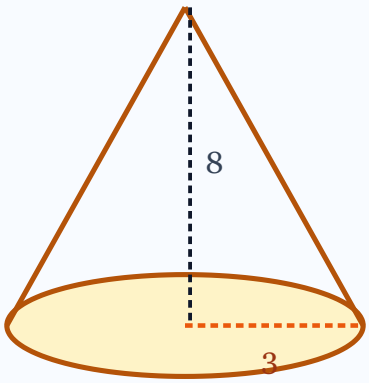
$$V = \frac{1}{3}Bh = \frac{1}{3}(256)(15) = \frac{3840}{3} = 1280.$$

Answer: 1280 cubic units.

Example 6: Cone

Find the volume of a cone with radius 3, height 8.

Solution



A cone fills exactly $\frac{1}{3}$ of the cylinder with the same base and height.

Identify the radius and height: $r = 3$, $h = 8$.

Apply the cone volume formula (one-third of the matching cylinder):

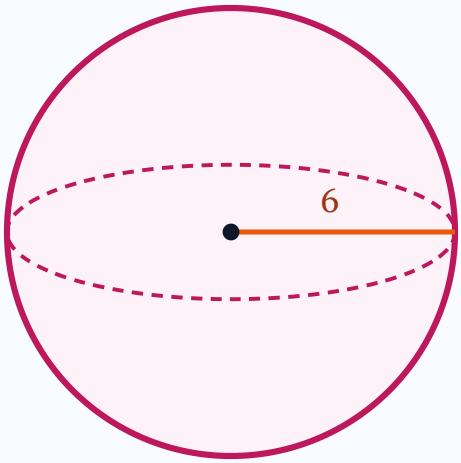
$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(9)(8) = \frac{72\pi}{3} = 24\pi.$$

Answer: 24π cubic units.

Example 7: Sphere

Find the volume of a sphere with radius 6.

Solution



$V = \frac{4}{3}\pi r^3$ uses the radius cubed.

Identify the radius: $r = 6$.

Apply the sphere volume formula:

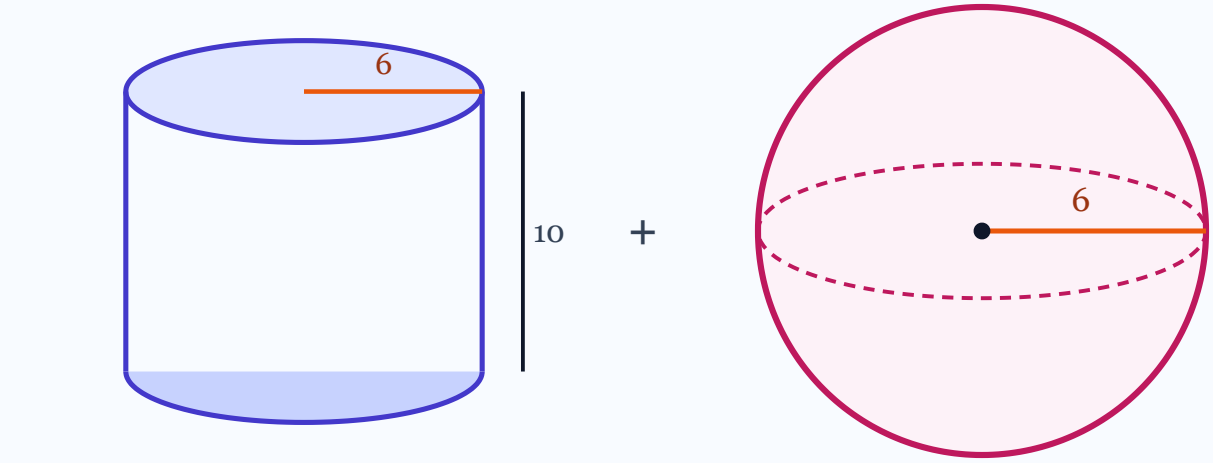
$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(216) = \frac{864\pi}{3} = 288\pi.$$

Answer: 288π cubic units.

Example 8: Composite solid — silo volume

A grain silo is a cylinder of radius 6 and height 10 topped by a hemisphere of the same radius. Find its total volume.

Solution



Total volume = cylinder volume + hemisphere volume (half a sphere).

Cylinder: $V_{\text{cyl}} = \pi r^2 h = \pi(36)(10) = 360\pi$.

Hemisphere (half a sphere of the same radius): $V_{\text{hemi}} = \frac{1}{2}\left(\frac{4}{3}\pi r^3\right) = \frac{2}{3}\pi(216) = 144\pi$.

Total: $V = 360\pi + 144\pi = 504\pi$.

Answer: 504π cubic units.

Example 9: Cone vs. cylinder — same base and height

A cone and a cylinder both have radius 5 and height 12. Find each volume and compare.

Solution

Cylinder: $V = \pi r^2 h = \pi(25)(12) = 300\pi$.

Cone: $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}(300\pi) = 100\pi$.

Comparison: the cone's volume is exactly $\frac{1}{3}$ of the cylinder's — this is true for *any* cone/cylinder pair sharing a base and height, not just this one.

Answers: cylinder $= 300\pi$; cone $= 100\pi$.

Example 10: Reverse problem — solve for a missing dimension

A cylinder has radius 5 and volume 250π. Find its height.

Solution

$V = \pi r^2 h \;\Rightarrow\; 250\pi = \pi(25)h$

Divide both sides by 25π: $h = \frac{250\pi}{25\pi} = 10$.

Check: $\pi(25)(10) = 250\pi$. ✓

Answer: $h = 10$.

Example 11: Word problem — capacity in liters

A cylindrical water tank has diameter 2 m and height 3 m. Find its volume in cubic meters, then convert to liters ($1\ \text{m}^3 = 1000$ L).

Solution

Radius: $r = \frac{2}{2} = 1$ m.

$V = \pi r^2 h = \pi(1)^2(3) = 3\pi\ \text{m}^3$ (exact form).

Using $\pi \approx 3.14159$: $V \approx 9.42\ \text{m}^3$.

Convert: $9.42\ \text{m}^3 \times 1000\frac{\text{L}}{\text{m}^3} \approx 9420$ L.

Answer: $3\pi\ \text{m}^3 \approx 9420$ L.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Rectangular prism

Find the volume of a rectangular prism $7 \times 4 \times 3$.

Answer: **Verification:**

Problem 2: Cylinder

Find the volume of a cylinder with radius 6, height 5.

Answer: **Verification:**

Problem 3: Pyramid (height given)

Square pyramid, base edge 9, height 4. Find volume.

Answer: **Verification:**

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Problem 4: Cone

Cone with radius 4, height 9. Find volume.

Answer: **Verification:**

Problem 5: Sphere from radius

Find the volume of a sphere with radius 9.

Answer: **Verification:**

Problem 6: Sphere from diameter

Find the volume of a sphere with diameter 8.

Answer: **Verification:**

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Problem 7: Cube

Find the volume of a cube with edge 8.

Answer: **Verification:**

Problem 8: Cylinder

Cylinder with radius 2, height 10. Find volume.

Answer: **Verification:**

Problem 9: Cone

Cone with radius 5, height 12. Find volume.

Answer: **Verification:**

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Problem 10: Pyramid — find h first

Square pyramid, base edge 10, slant height 13. Find the height, then the volume.

Answer: **Verification:**

Problem 11: Composite solid

A silo is a cylinder (radius 3, height 8) topped by a hemisphere of the same radius. Find the total volume.

Answer: **Verification:**

Problem 12: Reverse problem

A cone has radius 6 and volume 96π. Find its height.

Answer: **Verification:**

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Problem 13: Concept

A cone and a cylinder share the same base and height. What fraction of the cylinder's volume does the cone hold?

Answer: **Verification:**

Problem 14: Word problem

A cylindrical tank has diameter 6 m and height 4 m. Find its volume in m^3 (exact form), then estimate its capacity in liters.

Answer: **Verification:**

Problem 15: Challenge

A sphere is inscribed in a cube of edge length 10 so that it touches the center of all six faces (the sphere's radius equals half the cube's edge). Find the exact volume *inside the cube but outside the sphere*.

Answer: **Verification:**

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✓ Answer Key

1. 84
2. 180π
3. 108
4. 48π
5. 972π
6. $\frac{256\pi}{3}$
7. 512
8. 40π
9. 100π
10. $h = 12$; $V = 400$
11. 90π
12. $h = 8$
13. $\frac{1}{3}$
14. 36π m^3 $\approx$ 113,100 L
15. $1000 - \frac{500\pi}{3}$

Great work — keep practising!